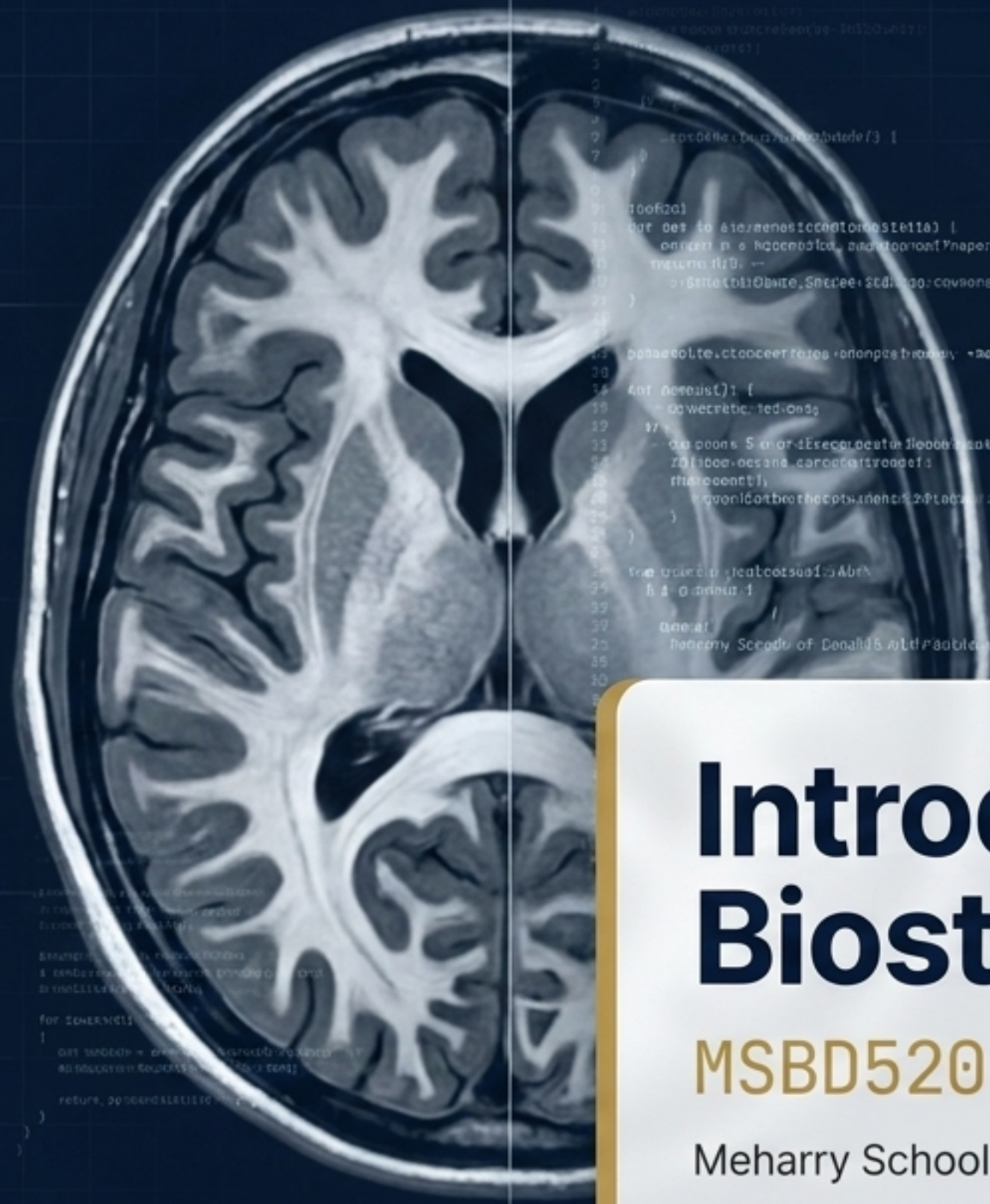




# Introduction to Biostatistics

MSBD520 | Fall 2026

Meharry School of Applied Computational Sciences

# Lead Investigator: Dr. Nazirah Mohd Khairi

## The Journey



Malaysia

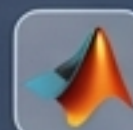


Nashville

> 10 years  
in Nashville  
(>15 total)

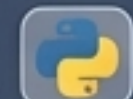
Loves  
Nature

## The Toolkit



MATLAB

> 10 years



Python

High proficiency



Bash

High proficiency



Octave

High proficiency



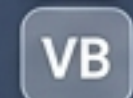
C++

>-5 years



Perl

1-3 years



VB

1-3 years

## Comms Protocol



3401 West End Ave,  
3rd floor



nazirah.mohdkhairi@mmc.edu  
(Direct email is the best way to reach me)



Office Hours:  
TBD via Blackboard

# The Analytical Staircase

## System Mastery

Variable selection, modeling techniques, regression diagnostics, and validation.

## Modeling

## Complexity

## Validation

## Application

## Foundation

**Mathematical Models:** Building phenomena across linear algebra, discrete math, and calculus.

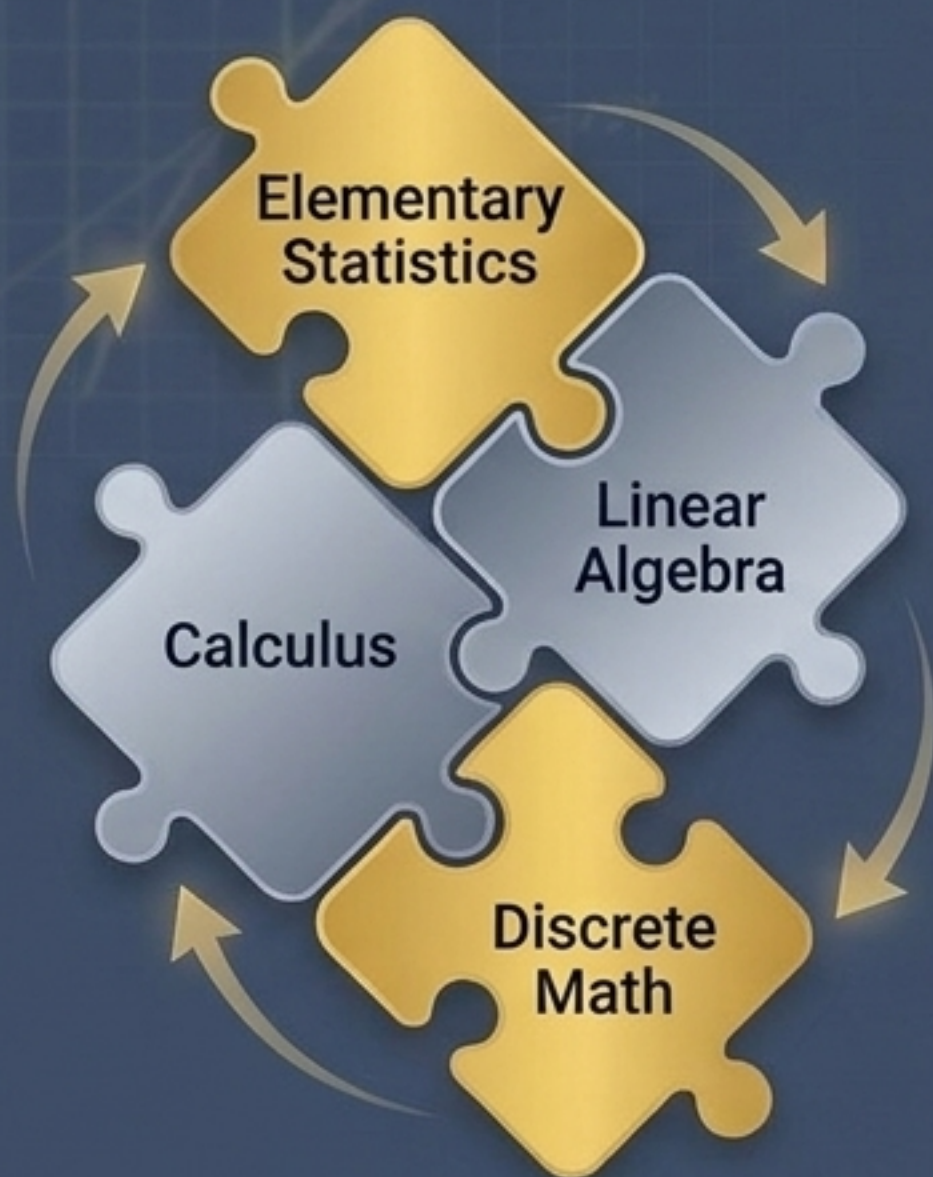
**Hypothesis Testing:** Parametric and nonparametric analysis methods.

**Study Design:** Applying descriptive and inferential methodologies.

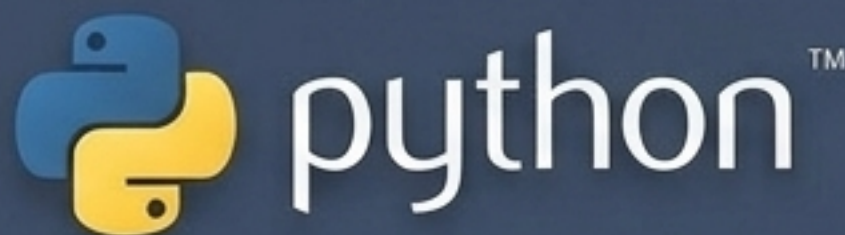
**Probability & Variation:** Basic concepts and standard statistical distributions.

# Equipment and Prerequisites

## Prerequisites

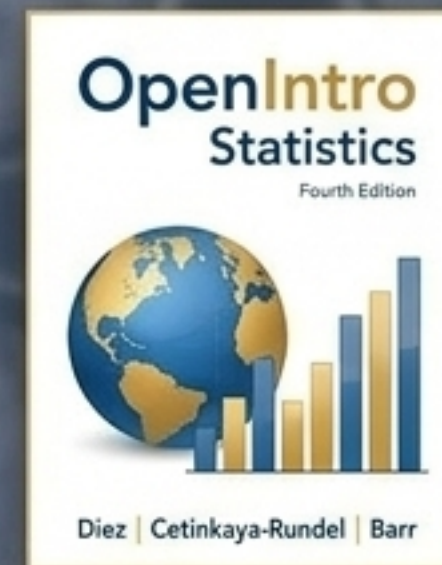


## Software Environment



Must be installed and familiarized for assignments and project development.

## The Library



Meharry Gold

### REQUIRED CORE:

OpenIntro Statistics, 4th Edition  
(Diez, Cetinkaya-Rundel, Barr)

### Reference Modules

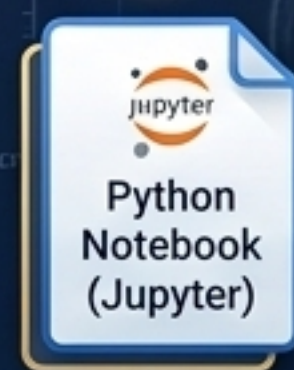
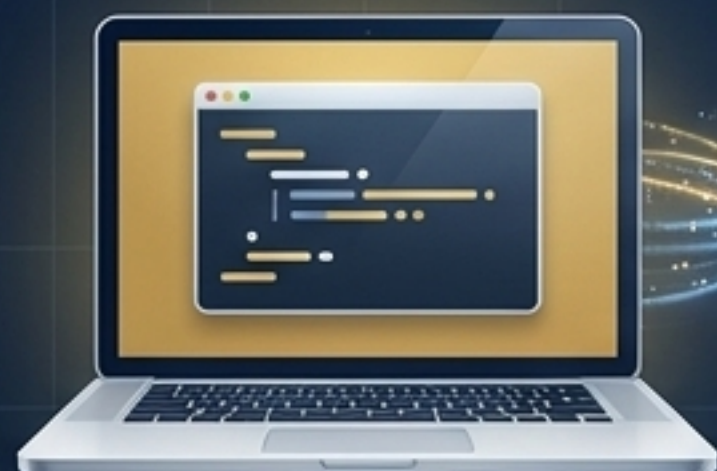
-  Introductory Statistics for Life/Biomedical Sciences (Vu/Harrington)
-  Fundamentals of Biostatistics 7th Ed (Rosner)
-  Think Stats 2nd Ed (Downey)

# Seamless Compute via Google Colab



The Advantage:  
No installation required.  
Operates entirely online.

Cloud Computing Bridge



## Diagnostic Box

How to check local Python:

MacOS (Terminal): `python3 --version >_`

Windows (CMD/PowerShell): `python3 --version >_`

# Navigating AI and Academic Integrity



## Allowed (Enhancement)

Grammar Check Tools, Writing Enhancement.



## Caution (The Core Rule)

All ideas, interpretations, and scholarly content must be solely by the student.

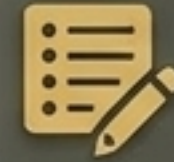
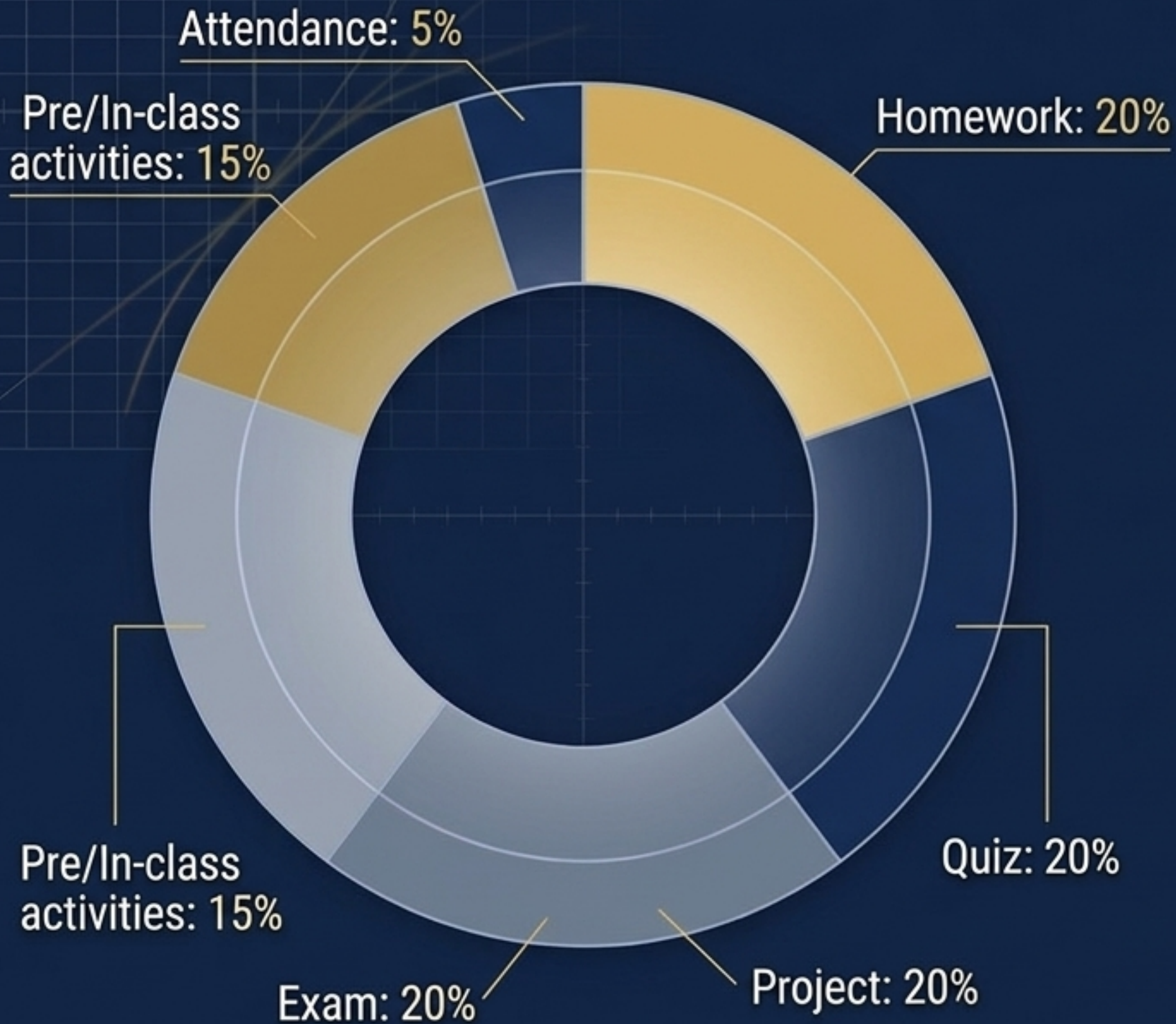


## Prohibited (Outsourcing)

Using AI to generate original scholarly thought or bypass interpretation.

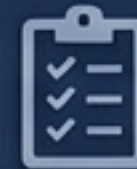
You are fully responsible for the accuracy, integrity, and originality of your submitted work.

# Grade Architecture and Assessment



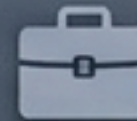
## Homework (20%)

Extensive exercises with interpretation questions.



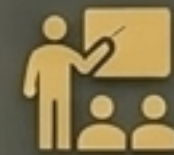
## Quizzes (20%)

Blackboard-based multiple choice/short answer on recent topics.



## Project & Exam (40%)

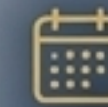
Major milestones; bigger tasks than standard HW; final exam due on the last day.



## In-Class (15%)

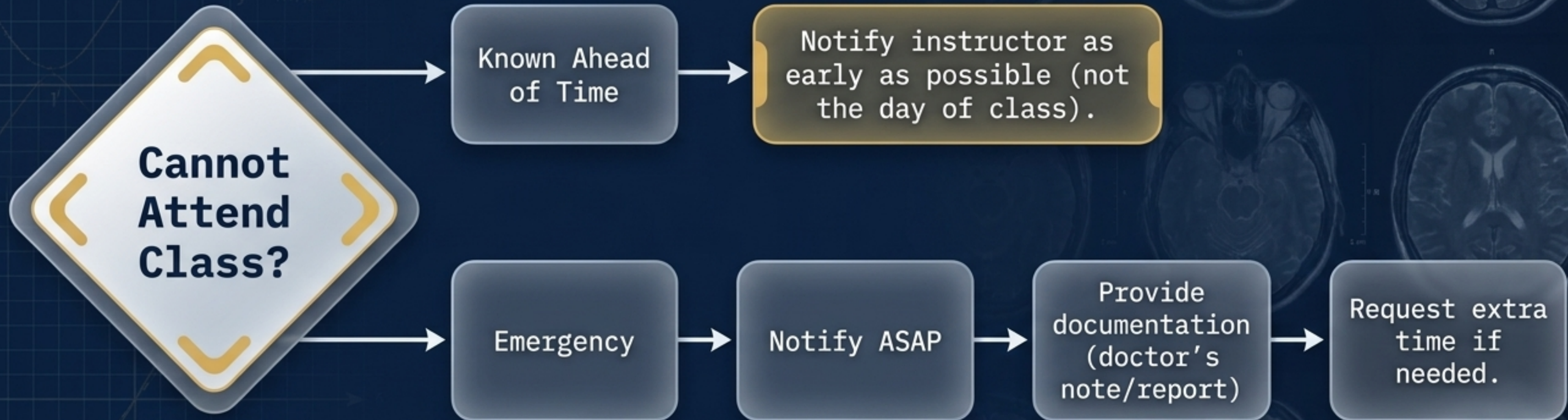
Individual or paired hands-on examples to learn stats by doing.

# Operational Protocols: Attendance and Deadlines



## Default State

Attendance is required  
(Thursdays 5:30pm–8:30pm CT). You  
are responsible for all deadlines.



## Missed Class Protocol

Check the recording, review slides, track assignment info and deadlines on Blackboard.

# The Analytical Terrain: Navigating Uncertainty

Probability



A mathematical language used to discuss uncertain events

## Statistics

The science concerned with developing and studying methods for collecting, analyzing, interpreting, and presenting empirical data.

## Biostatistics

The application of these statistical principles in medicine, public health, biology, and life sciences.

# The Blueprint of Inference: Two Branches of Statistics

Descriptive

Statistics

Inferential

## Descriptive Statistics

Action

Summarizing and communicating information.

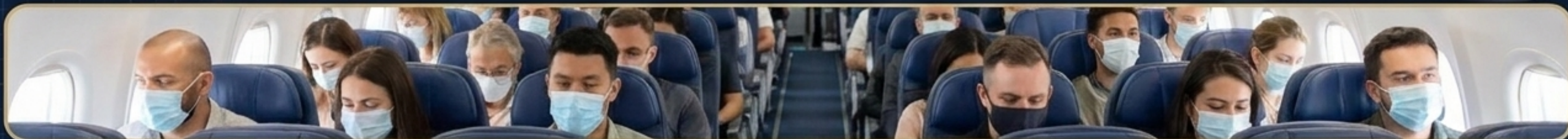
Scope

Refers ONLY to the actual data available.

## Inferential Statistics

Drawing conclusions and making predictions.

Goes BEYOND the immediate data collected.

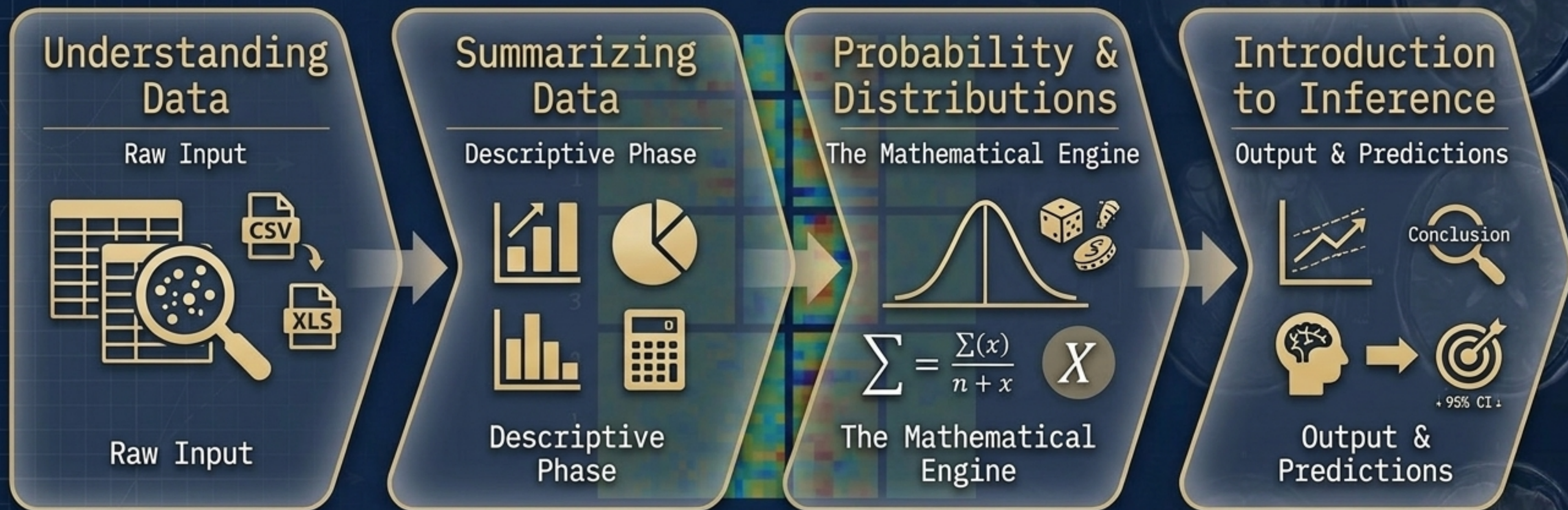


The  
Airline  
Example

What is the average age of the passengers on THIS specific flight?

Based on this flight, what is the predicted average age of ALL millions of US airline passengers?

# The Biostatistician's Workflow



This is the sequential path we will take to build our statistical capabilities over the semester.

# Immediate Action Items



**Systems Check:** Familiarize yourself with Blackboard, the Class App, and the Course Syllabus.



**Material Prep:** Access the OpenIntro Statistics Book and website.



**Environment Check:** Boot up your Python/R/Programming platform (or Google Colab) to ensure readiness.



**The First Milestone:** Submit In-Class 1 (Deadline: Aug 19).

## Icebreaker Callout Box

**Cohort Calibration: Let's get to know each other**

Be prepared to share:

- Your background & current city
- Non-academic projects
- Summer fun facts
- Research & career interests